Primus-Turbo 2026 Q1 Summary

1.GEMM

AMD docker image: docker.io/rocm/primus:v26.1

NVIDIA docker image: nvcr.io/nvidia/nemo:26.02

GEMM-BF16	MI355 ROCm-TE	GB200 NV-TE*	MI355 Turbo Hipblast	MI355 Turbo CK	MI355 Turbo Triton	MI355 Turbo AutoTune
Avg-Forward	1524.76702	1434.365	1534.634	N/A	1146.94	1532.047
Avg-Backward	1339.428	1310.245	1350.293	N/A	1225.549	1351.097
GEMM-FP8-Tensorwise	MI355 ROCm-TE	GB200 NV-TE*	MI355 Turbo Hipblast	MI355 Turbo CK	MI355 Turbo Triton	MI355 Turbo AutoTune
Avg-Forward	1965.585	2081.86	2297.809	1675.359	1878.135	2295.154
Avg-Backward	2253.957	2204.07	1354.071	160.4699	1884.492	1866.042
GEMM-FP8-Blockwise	MI355	GB200	MI355 Turbo	MI355 Turbo	MI355 Turbo	MI355 Turbo

	ROCm-TE	NV-TE*	Hipblast	CK	Triton	AutoTune
Avg-Forward	N/A	1944.059	N/A	474.1665	1369.906	1350.773
Avg-Backward	N/A	2062.787	N/A	227.4014	1009.11	1006.693
GEMM-MXFP8*	**MI355 ROCm-TE**	**GB200 NV-TE***	**MI355 Turbo Hipblast**	**MI355 Turbo CK**	**MI355 Turbo Triton**	**MI355 Turbo AutoTune**
Avg-Forward	N/A	1947.08	1010.546	N/A	N/A	1011.993
Avg-Backward	N/A	1864.25	1119.425	N/A	N/A	1121.101
GEMM-MXFP4*	**MI355 ROCm-TE**	**GB200 NV-TE***	**MI355 Turbo Hipblast**	**MI355 Turbo Aiter**	**MI355 Turbo Triton**	**MI355 Turbo AutoTune**
Avg-Forward	N/A	2127.23	1723.193	2346.74	N/A	1736.623
Avg-Backward	N/A	2390.83	2043.355	2490.23	N/	2035.219

NOTE:

1. The workload of MXFP4 and MXFP8 only include llama2-7B.

2. Grouped GEMM

NVIDIA docker image: nvcr.io/nvidia/nemo:26.02

Grouped-GEMM-BF16	**MI355 ROCm-TE**	**GB200 NV-TE***	**MI355 Turbo Hipblast**	**MI355 Turbo CK**	**MI355 Turbo Triton**	**MI355 Turbo AutoTune**
Avg-Forward	904.39	1133.233	940.09	1065.57	1098.73	1084.42
Avg-Backward	501.08	931.9719	659.19	746.32	922.26	905.16
Grouped-GEMM-FP8 Tensorwise	**MI355 ROCm-TE**	**GB200 NV-TE***	**MI355 Turbo Hipblast**	**MI355 Turbo CK**	**MI355 Turbo Triton**	**MI355 Turbo AutoTune**
Avg-Forward	481.42	992.6915	1085.80	720.19	1198.73	1203.07
Avg-Backward	625.00	1223.462	642.81	93.62	1304.85	1301.38
Grouped-GEMM-FP8 Rowwise	**MI355 ROCm-TE**	**GB200 NV-TE***	**MI355 Turbo Hipblast**	**MI355 Turbo CK**	**MI355 Turbo Triton**	**MI355 Turbo AutoTune**
Avg-Forward	N/A	N/A	N/A	430.05	859.74	852.30
Avg-Backward	N/A	N/A	N/A	87.80	1179.64	1167.21
Grouped-GEMM-FP8 Blockwise	**MI355 ROCm-TE**	**GB200 NV-TE***	**MI355 Turbo Hipblast**	**MI355 Turbo CK**	**MI355 Turbo Triton**	**MI355 Turbo AutoTune**
Avg-Forward	N/A	968.7614	N/A	296.65	653.49	649.08
Avg-Backward	N/A	1112.789	N/A	202.27	556.95	555.17

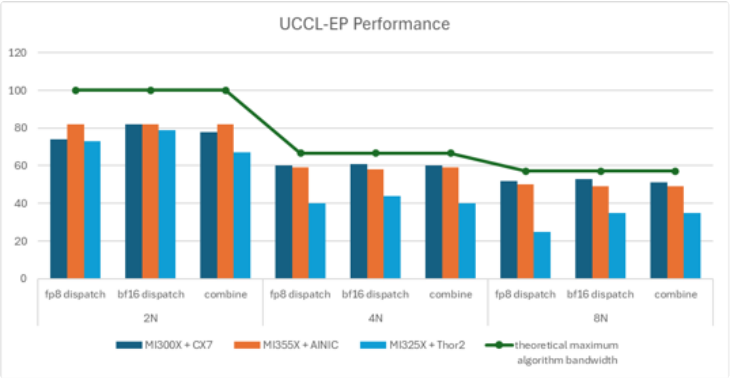
raw data：https://amdcloud-my.sharepoint.com/:f:/g/personal/ruibzhan_amd_com/IgAuC4F52HskQo9UZnbX3mByAZJkBbvjE6RsUxfZUL2wyQY?e=7ihTm9 Connect your OneDrive account

3.DeepEP

Test Configuration:

- Device: MI300X
- Tokens: 4096
- Hidden Dimension: 7168
- TopK: 8
- Router: Random Uniform
- Local Experts: 8
- Ranks: 16/32/64

		MI300X-UCCL-EP	H800-DeepSeek/DeepEP	speed up
2N	FP8 Dispatch	74	43	1.72
	Combine	78	43	1.81
4N	FP8 Dispatch	60	58	1.03
	Combine	60	57	1.05
8N	FP8 Dispatch	52	51	1.02
	Combine	51	50	1.02



UCCL-EP performance on CX7, AINIC and Thor2

4.Primus E2E Test